PAL API Reference

Generated from public `Protocol` declarations and top-level factory signatures in `NativeCore.py`. Regenerate with `python Documentation/generators/generate_api_reference.py`; use `--check` to detect drift. These declarations describe the annotation/autocomplete surface; consult API Guide for behavior and the implementation/tests for runtime semantics.

Factories and utilities

```
FastMode()
# Select PAL fast mode before constructing any PAL object or window; the process-wide mode cannot be
 changed after construction.
NewIList()
# Create an empty PAL integer list usable from Python and supported compiled functions.
MooreHood(dim, excludeCenter=False)
# Build relative integer offsets for a Moore neighborhood in `dim` dimensions; optionally exclude th
e center offset.
VonNeumannHood(dim, excludeCenter=False)
# Build relative integer offsets for a Von Neumann neighborhood in `dim` dimensions; optionally excl
ude the center offset.
CircleHood(dim, rad, excludeCenter=False)
# Build relative integer offsets within radius `rad` in `dim` dimensions; optionally exclude the cen
ter offset.
NewMultinomial(other=None)
# Create a PAL multinomial/binomial sampler, optionally reusing configuration from `other`.
NewGrid(dimensions, dtype)
# Create a typed lattice grid with the requested dimensions and element dtype.
NewAgentGrid(dimensions, numAgentProps=0, isStackable=False)
# Create a grid for individual agents; `numAgentProps` sets the number of per-agent properties and `
isStackable` controls whether multiple agents may occupy a site.
NewPopGrid(dimensions, capacity=None)
# Create a lattice of integer population counts, optionally limiting the count at each site with `ca
pacity`.
NewPDEgrid(dimensions)
# Create a lattice-backed scalar field for numerical transport operations.
```

Multinomial

Kind	Declaration	Description
method	`Binomial(self, n: int, p: float) -> int`	Description not yet provided.
method	`Setup(self, n: int)`	Configure the sampler for n trials before repeated sampling.
method	`Sample(self, p: float) -> int`	Description not yet provided.

IList

Kind	Declaration	Description
method	`__len__(self) -> int`	Description not yet provided.
method	`__getitem__(self, index: int) -> int`	Description not yet provided.
method	`Append(self, i: int)`	Append integer `i` to the end of the list.
method	`Clear(self)`	Remove all values from the list.
method	`Random(self) -> int`	Description not yet provided.
method	`Shuffle(self)`	Shuffle the list in place using PAL's random stream.
method	`All(self) -> np.ndarray`	Description not yet provided.
method	`Iter(self) -> np.ndarray`	Description not yet provided.

AgentGrid

Kind	Declaration	Description
method	`__len__(self) -> int`	Description not yet provided.
property	`xDim(self) -> int`	Description not yet provided.
property	`yDim(self) -> int`	Description not yet provided.
property	`zDim(self) -> int`	Description not yet provided.
property	`nDims(self) -> int`	Description not yet provided.
property	`wrapX(self) -> bool`	Description not yet provided.
property	`wrapY(self) -> bool`	Description not yet provided.
property	`wrapZ(self) -> bool`	Description not yet provided.
method	`Box(self, x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None)`	Iterate lattice coordinates in the half-open rectangular region [x1, x2) x [y1, y2) x [z1, z2); omit unused axes in lower dimensions.
method	`Hood(self, hood, x: int, y: int=None, z: int=None, *, unroll: bool=False)`	Apply relative neighborhood offsets at a lattice coordinate, respecting wrapped and nonwrapped boundaries. `unroll=True` enables compiled-loop expansion where supported.
method	`GetPop(self) -> int`	Description not yet provided.
method	`Alive(self, agent: int) -> bool`	Description not yet provided.
method	`NewAgentSQ(self, x: int, y: int=-1, z: int=-1) -> int`	Description not yet provided.
method	`NewAgent(self, x: float, y: float=-1.0, z: float=-1.0) -> int`	Description not yet provided.
method	`Dispose(self, agent: int)`	Remove a living agent from the grid; its handle must not be used as a live agent afterward.
method	`I(self, agent: int) -> int`	Description not yet provided.
method	`XSQ(self, agent: int) -> int`	Description not yet provided.
method	`YSQ(self, agent: int) -> int`	Description not yet provided.
method	`ZSQ(self, agent: int) -> int`	Description not yet provided.
method	`X(self, agent: int) -> float`	Description not yet provided.
method	`Y(self, agent: int) -> float`	Description not yet provided.
method	`Z(self, agent: int) -> float`	Description not yet provided.
method	`__getitem__(self, key) -> float`	Description not yet provided.
method	`__setitem__(self, key, value: float)`	Set a supported per-agent property value for the selected agent/property key.
method	`ToI(self, x: int, y: int=-1, z: int=-1) -> int`	Description not yet provided.

Kind	Declaration	Description
method	`ItoX(self, i: int) -> int`	Description not yet provided.
method	`ItoY(self, i: int) -> int`	Description not yet provided.
method	`ItoZ(self, i: int) -> int`	Description not yet provided.
method	`InWrapSQX(self, value: int) -> int`	Description not yet provided.
method	`InWrapSQY(self, value: int) -> int`	Description not yet provided.
method	`InWrapSQZ(self, value: int) -> int`	Description not yet provided.
method	`InWrapX(self, value: float) -> float`	Description not yet provided.
method	`InWrapY(self, value: float) -> float`	Description not yet provided.
method	`InWrapZ(self, value: float) -> float`	Description not yet provided.
method	`DispWrapX(self, x1: float, x2: float) -> float`	Description not yet provided.
method	`DispWrapY(self, y1: float, y2: float) -> float`	Description not yet provided.
method	`DispWrapZ(self, z1: float, z2: float) -> float`	Description not yet provided.
property	`counts(self) -> object`	Description not yet provided.
method	`LastAgent(self, x: int, y: int=-1, z: int=-1) -> int`	Description not yet provided.
method	`MoveSQ(self, agent: int, x: int, y: int=-1, z: int=-1)`	Move an agent to a lattice site specified by integer coordinates or a supported linear site index.
method	`Move(self, agent: int, x: float, y: float=-1.0, z: float=-1.0)`	Move an agent to continuous coordinates; use `MoveSQ` when movement should target a lattice site.
method	`AgentsAt(self, x: int, y: int=-1, z: int=-1)`	Return the agent handle or handles occupying the specified lattice site.
method	`AgentsInRadius(self, rad: float, x: float, y: float=-1.0, z: float=-1.0, *, exclude: int=None)`	Return agents within radius `rad` of a continuous position; `exclude` can omit one agent from the results.
method	`All(self, shuffle: bool=False) -> np.ndarray`	Description not yet provided.

Grid

Kind	Declaration	Description
method	`__len__(self) -> int`	Description not yet provided.
method	`__getitem__(self, key)`	Read the value or state selected by the supported index or coordinate key.
method	`__setitem__(self, key, value)`	Assign the value or state selected by the supported index or coordinate key.
property	`xDim(self) -> int`	Description not yet provided.
property	`yDim(self) -> int`	Description not yet provided.
property	`zDim(self) -> int`	Description not yet provided.
property	`nDims(self) -> int`	Description not yet provided.
property	`wrapX(self) -> bool`	Description not yet provided.
property	`wrapY(self) -> bool`	Description not yet provided.
property	`wrapZ(self) -> bool`	Description not yet provided.
method	`Box(self, x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None)`	Iterate lattice coordinates in the half-open rectangular region [x1, x2) x [y1, y2) x [z1, z2); omit unused axes in lower dimensions.
method	`Hood(self, hood, x: int, y: int=None, z: int=None, *, unroll: bool=False)`	Apply relative neighborhood offsets at a lattice coordinate, respecting wrapped and nonwrapped boundaries. `unroll=True` enables compiled-loop expansion where supported.
method	`ToI(self, x: int, y: int=-1, z: int=-1) -> int`	Description not yet provided.
method	`ItoX(self, i: int) -> int`	Description not yet provided.
method	`ItoY(self, i: int) -> int`	Description not yet provided.
method	`ItoZ(self, i: int) -> int`	Description not yet provided.
method	`InWrapX(self, value: int) -> int`	Description not yet provided.
method	`InWrapY(self, value: int) -> int`	Description not yet provided.
method	`InWrapZ(self, value: int) -> int`	Description not yet provided.

PopGrid

Kind	Declaration	Description
method	`__len__(self) -> int`	Description not yet provided.
property	`xDim(self) -> int`	Description not yet provided.
property	`yDim(self) -> int`	Description not yet provided.
property	`zDim(self) -> int`	Description not yet provided.
property	`nDims(self) -> int`	Description not yet provided.
property	`wrapX(self) -> bool`	Description not yet provided.
property	`wrapY(self) -> bool`	Description not yet provided.
property	`wrapZ(self) -> bool`	Description not yet provided.
method	`Box(self, x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None)`	Iterate lattice coordinates in the half-open rectangular region [x1, x2) x [y1, y2) x [z1, z2); omit unused axes in lower dimensions.
method	`Hood(self, hood, x: int, y: int=None, z: int=None, *, unroll: bool=False)`	Apply relative neighborhood offsets at a lattice coordinate, respecting wrapped and nonwrapped boundaries. `unroll=True` enables compiled-loop expansion where supported.
method	`ToI(self, x: int, y: int=-1, z: int=-1) -> int`	Description not yet provided.
method	`ItoX(self, i: int) -> int`	Description not yet provided.
method	`ItoY(self, i: int) -> int`	Description not yet provided.
method	`ItoZ(self, i: int) -> int`	Description not yet provided.
method	`InWrapX(self, value: int) -> int`	Description not yet provided.
method	`InWrapY(self, value: int) -> int`	Description not yet provided.
method	`InWrapZ(self, value: int) -> int`	Description not yet provided.
method	`__getitem__(self, key) -> int`	Description not yet provided.
method	`__setitem__(self, key, value: int)`	Set the population count at the selected site immediately; this is distinct from queueing a delta with `Add()`.
method	`Add(self, value: int, x: int, y: int=-1, z: int=-1)`	Queue a signed population-count delta at a site; call `Update()` to apply pending deltas.
method	`Update(self)`	Apply queued population-count deltas to the current counts.
method	`Reset(self)`	Clear population counts and any pending deltas.
method	`GetPop(self) -> int`	Description not yet provided.
method	`All(self) -> np.ndarray`	Description not yet provided.

PDEgrid

Kind	Declaration	Description
method	`__len__(self) -> int`	Description not yet provided.
property	`xDim(self) -> int`	Description not yet provided.
property	`yDim(self) -> int`	Description not yet provided.
property	`zDim(self) -> int`	Description not yet provided.
property	`nDims(self) -> int`	Description not yet provided.
property	`wrapX(self) -> bool`	Description not yet provided.
property	`wrapY(self) -> bool`	Description not yet provided.
property	`wrapZ(self) -> bool`	Description not yet provided.
method	`Box(self, x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None, z2: int=None)`	Iterate lattice coordinates in the half-open rectangular region [x1, x2) x [y1, y2) x [z1, z2); omit unused axes in lower dimensions.
method	`Hood(self, hood, x: int, y: int=None, z: int=None, *, unroll: bool=False)`	Apply relative neighborhood offsets at a lattice coordinate, respecting wrapped and nonwrapped boundaries. `unroll=True` enables compiled-loop expansion where supported.
method	`ToI(self, x: int, y: int=-1, z: int=-1) -> int`	Description not yet provided.
method	`ItoX(self, i: int) -> int`	Description not yet provided.
method	`ItoY(self, i: int) -> int`	Description not yet provided.
method	`ItoZ(self, i: int) -> int`	Description not yet provided.
method	`InWrapX(self, value: int) -> int`	Description not yet provided.
method	`InWrapY(self, value: int) -> int`	Description not yet provided.
method	`InWrapZ(self, value: int) -> int`	Description not yet provided.
method	`__getitem__(self, key) -> float`	Description not yet provided.
method	`__setitem__(self, key, value: float)`	Set the field value at the selected site immediately; this is distinct from queueing a change with `Add()`.
method	`Add(self, value: float, x: int, y: int=-1, z: int=-1)`	Queue a field-value delta at a site; call `Update()` to apply pending changes.
method	`Update(self)`	Apply queued field changes to the current field.
method	`Reset(self)`	Clear the field and any pending changes.
method	`SetTimeSpaceStep(self, dt: float, dx: float, dy: float=1.0, dz: float=1.0)`	Set the temporal step `dt` and spatial steps `dx`, `dy`, and `dz` used by numerical transport methods.
method	`Dx(self) -> float`	Description not yet provided.

Kind	Declaration	Description
method	`Dy(self) -> float`	Description not yet provided.
method	`Dz(self) -> float`	Description not yet provided.
method	`Dt(self) -> float`	Description not yet provided.
method	`Diffusion(self, rateConstant: float, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply scalar-coefficient diffusion using the configured grid spacing and optional per-face boundary conditions.
method	`DiffusionMask(self, rateConstant: float, mask=None)`	Apply diffusion only across interfaces allowed by `mask`.
method	`DiffusionField(self, rateConstants, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply diffusion with spatially varying rate constants and optional boundary conditions.
method	`DiffusionInterfaces(self, rateConstantsX, rateConstantsY=None, rateConstantsZ=None, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply diffusion using rate constants defined on grid interfaces; separate arrays may specify x, y, and z interfaces.
method	`DiffusionADI(self, rateConstant: float, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply diffusion using an alternating-direction implicit method.
method	`DiffusionRadialCircle(self, rateConstant: float, outerBC=None)`	Apply radial diffusion on a circular domain with an optional outer boundary condition.
method	`DiffusionRadialSphere(self, rateConstant: float, outerBC=None)`	Apply radial diffusion on a spherical domain with an optional outer boundary condition.
method	`Advection(self, vx: float, vy: float=0.0, vz: float=0.0, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply advection using a constant velocity vector and optional per-face boundary conditions.
method	`AdvectionField(self, xVels, yVels=None, zVels=None, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply advection using spatially varying velocity fields.
method	`AdvectionInterfaces(self, xVels, yVels=None, zVels=None, xMinBC=None, xMaxBC=None, yMinBC=None, yMaxBC=None, zMinBC=None, zMaxBC=None)`	Apply advection using velocity values defined on grid interfaces.

PixWindow visualization

```
StartPixWindow(xDim, yDim, scale=1, title='PAL', headless=False) -> tuple[Pix, PixWindow]
```

Description not yet provided.

Kind	Declaration	Description
method	`IsOpen(self) -> bool`	Description not yet provided.
method	`Update(self)`	Description not yet provided.
method	`Save(self, path: str, block=False)`	Description not yet provided.
method	`StartGif(self, path: str, delay=100)`	Description not yet provided.
method	`AddGifFrame(self, block=False)`	Description not yet provided.
method	`StopGif(self)`	Description not yet provided.
method	`Close(self)`	Description not yet provided.

OpenGLWindow visualization

```
StartOpenGLWindow(xDim, yDim, zDim=None, width=800, height=800, title='PAL', headless=False) -> tupl
e[OpenGLDraw, OpenGLWindow]
```

Description not yet provided.

Kind	Declaration	Description
method	`Update(self)`	Description not yet provided.
method	`IsOpen(self) -> bool`	Description not yet provided.
method	`StartGif(self, path, delay=100)`	Description not yet provided.
method	`AddGifFrame(self, block=False, timeout=30)`	Description not yet provided.
method	`StopGif(self, timeout=30)`	Description not yet provided.
method	`Circle(self, rad, color, x, y, z=None)`	Description not yet provided.
method	`Box(self, xLen, color, x, y, z=None, yLen=None, zLen=None)`	Description not yet provided.
method	`BoxSQ(self, color, x, y, z=None)`	Description not yet provided.
method	`Line(self, width, color, x1, y1, x2, y2, z1=None, z2=None)`	Description not yet provided.
method	`Borders(self, width, color)`	Description not yet provided.
method	`Camera(self, x, y, z, yaw=None, pitch=None)`	Description not yet provided.
method	`Clear(self)`	Description not yet provided.
method	`Background(self, color)`	Description not yet provided.

Kind	Declaration	Description
method	`Save(self, path: str, block=False)`	Description not yet provided.
method	`Close(self)`	Description not yet provided.

The `OpenGLWindow` table follows its source Protocol, which also contains method implementations; repeated declarations are listed once. `OpenGLDraw` is a compiled primitive collector, while the returned window handles rendering and output. Consult the API Guide for which object to use in Python and compiled drawing code.

Pix drawing buffer

Kind	Declaration	Description
method	`__setitem__(self, key, color)`	Description not yet provided.
method	`Xdim(self)`	Description not yet provided.
method	`Ydim(self)`	Description not yet provided.
method	`ToI(self, x, y)`	Description not yet provided.
method	`ItoX(self, i)`	Description not yet provided.
method	`ItoY(self, i)`	Description not yet provided.
method	`__len__(self)`	Description not yet provided.

OpenGLDraw primitive collector

Kind	Declaration	Description
method	`Clear(self)`	Description not yet provided.
method	`Circle(self, rad, color, x, y, z=0.0)`	Description not yet provided.
method	`Box(self, xLen, color, x, y, z=0.0, yLen=-1.0, zLen=-1.0)`	Description not yet provided.
method	`BoxSQ(self, color, x, y, z=0.0)`	Description not yet provided.
method	`Line(self, width, color, x1, y1, x2, y2, z1=0.0, z2=0.0)`	Description not yet provided.

Scope and validation

This reference lists public Protocol signatures, not internal safe/fast jitclass methods. A signature does not capture every runtime overload, indexing form, or validation rule. See the API Guide and executable tests. Visualization constructors and window Protocol methods are included; Pix buffer and OpenGLDraw method signatures are also included; consult the API Guide for behavior.